

JEE Mains 2019 Chapter wise Question Bank

Binomial Theorem – Questions

Q1

The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$

- (1) 14 (2) 15
(3) 10 (4) 12

9 Jan Evening

Q2

If $\sum_{i=1}^{20} \left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} \right)^3 = \frac{k}{21}$, then k equals:

- (1) 400 (2) 50
(3) 200 (4) 100

10 Jan Morning

Q3

If the third term in the binomial expansion of $\left(1 + x^{\log_2 x}\right)^5$ equals 2560, then a possible value of x is:

- (1) $\frac{1}{4}$ (2) $4\sqrt{2}$ (3) $\frac{1}{8}$ (4) $2\sqrt{2}$

10 Jan Morning

Q4

The positive value of λ for which the co-efficient of x^2 in

the expression $x^2 \left(\sqrt{x} + \frac{\lambda}{x^2} \right)^{10}$ is 720, is:

- (1) 4 (2) $2\sqrt{2}$
(3) $\sqrt{5}$ (4) 3

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Q5

If $\sum_{r=0}^{25} \left\{ {}^{50}C_r \cdot {}^{50-r}C_{25-r} \right\} = K \left({}^{50}C_{25} \right)$, then K is equal to:

- (1) $(25)^2$ (2) $2^{25} - 1$
(3) 2^{24} (4) 2^{25}

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Q6

The value of r for which

${}^{20}C_r {}^{20}C_0 + {}^{20}C_{r-1} {}^{20}C_1 + {}^{20}C_{r-2} {}^{20}C_2 + \dots + {}^{20}C_0 {}^{20}C_r$ is maximum, is:

- (1) 15 (2) 20
(3) 11 (4) 10

11 Jan Morning

Q7

The sum of the real values of x for which the middle term in

the binomial expansion of $\left(\frac{x^3}{3} + \frac{3}{x} \right)^8$ equals 5670 is:

- (1) 0 (2) 6
(3) 4 (4) 8

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Q8

Binomial Theorem

Let $S_n = 1 + q + q^2 + \dots + q^n$ and

$$T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$$

where q is a real number and $q \neq 1$. If

${}^{101}C_1 + {}^{101}C_2 \cdot S_1 + \dots + {}^{101}C_{101} \cdot S_{100} = \alpha T_{100}$, then α is equal to :

- (1) 2^{99} (2) 202
(3) 200 (4) 2^{100}

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Q9

Let $(x+10)^{50} + (x-10)^{50} = a_0 + a_1x + a_2x^2 + \dots + a_{50}x^{50}$, for

all $x \in \mathbf{R}$; then $\frac{a_2}{a_0}$ is equal to :

- (1) 12.50 (2) 12.00
(3) 12.25 (4) 12.75

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Q10

A ratio of the 5th term from the beginning to the 5th term from

the end in the binomial expansion of $\left(2^{\frac{1}{3}} + \frac{1}{2(3)^{\frac{1}{3}}}\right)^{10}$ is:

- (1) $1:2(6)^{\frac{1}{3}}$ (2) $1:4(16)^{\frac{1}{3}}$
(3) $4(36)^{\frac{1}{3}}:1$ (4) $2(36)^{\frac{1}{3}}:1$

12 Jan Morning

Q11

The total number of irrational terms in the binomial expansion

of $\left(7^{\frac{1}{5}} - 3^{\frac{1}{10}}\right)^{60}$ is :

- (1) 55 (2) 49
(3) 48 (4) 54

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Q12

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The sum of the co-efficients of all even degree terms in x in the expansion of $\left(x + \sqrt{x^3 - 1}\right)^6 + \left(x - \sqrt{x^3 - 1}\right)^6$, ($x > 1$) is equal to :

- (1) 29 (2) 32 (3) 26 (4) 24

8 April Morning

Q13

The sum of the series $2 \cdot {}^{20}C_0 + 5 \cdot {}^{20}C_1 + 8 \cdot {}^{20}C_2 + 11 \cdot {}^{20}C_3 + \dots + 62 \cdot {}^{20}C_{20}$ is equal to :

- (1) 2^{26} (2) 2^{25} (3) 2^{23} (4) 2^{24}

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Q14

If the fourth term in the binomial expansion of $\left(\sqrt{\frac{1}{x^{1+\log_{10} x}}} + x^{\frac{1}{12}}\right)^6$ is equal to 200, and $x > 1$, then the

value of x is:

- (1) 100 (2) 10 (3) 10^3 (4) 10^4

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Q15

If the fourth term in the Binomial expansion of $\left(\frac{2}{x} + x^{\log_8 x}\right)^6$ ($x > 0$) is 20×8^7 , then a value of x is:

- (1) 8^3 (2) 8^2 (3) 8 (4) 8^{-2}

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Q16

If some three consecutive coefficients in the binomial expansion of $(x+1)^n$ in powers of x are in the ratio 2:15:70, then the average of these three coefficients is:

- (1) 964 (2) 232 (3) 227 (4) 625

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Q17

If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1-3x)^{15}$ in powers of x , then the ordered pair (a, b) is equal to :

- (1) $(28, 861)$ (2) $(-54, 315)$
(3) $(28, 315)$ (4) $(-21, 714)$

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Q18

The smallest natural number n , such that the coefficient of x in the expansion of $\left(x^2 + \frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is :

- (1) 38 (2) 58 (3) 23 (4) 35

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Q19

The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is :

- (1) 84 (2) -126 (3) -84 (4) 126

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Q20

If ${}^{20}C_1 + (2^2) {}^{20}C_2 + (3^2) {}^{20}C_3 + \dots + (20^2) {}^{20}C_{20} = A(2^\beta)$, then the ordered pair (A, β) is equal to :

- (1) (420, 19) (2) (420, 18)
(3) (380, 18) (4) (380, 19)

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Q21

The term independent of x in the expansion of

$$\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$$
 is equal to :

- (1) -72 (2) 36 (3) -36 (4) -108

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Binomial Theorem - Solutions

Q1

(2) Consider the expression

$$\begin{aligned} \left(\frac{1-t^6}{1-t}\right)^3 &= (1-t^6)^3(1-t)^{-3} \\ &= (1-3t^6+3t^{12}-t^{18})\left(1+3t+\frac{3\cdot 4}{2!}t^2\right. \\ &\quad \left.+\frac{3\cdot 4\cdot 5}{3!}t^3+\frac{3\cdot 4\cdot 5\cdot 6}{4!}t^4+\dots\right) \end{aligned}$$

Hence, the coefficient of $t^4 = 1 \cdot \frac{3\cdot 4\cdot 5\cdot 6}{4!}$

$$\begin{aligned} &= \frac{3\times 4\times 5\times 6}{4\times 3\times 2\times 1} \\ &= 15 \end{aligned}$$

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Q2

(4) Consider the expression, $\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} = \frac{{}^{20}C_{i-1}}{{}^{21}C_1}$

$$= \frac{20!}{(i-1)!(21-i)!} \times \frac{i!(21-i)!}{21!} = \frac{i}{21}$$

$$\begin{aligned} \therefore \sum_{i=1}^{20} \left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} \right)^3 &= \sum_{i=1}^{20} \left(\frac{i}{21} \right)^3 = \frac{(1)}{(21)^3} \sum_{i=1}^{20} i^3 \\ &= \frac{1}{(21)^3} \times \left(\frac{20 \times 21}{2} \right)^2 = \frac{100}{21} \end{aligned}$$

$$\therefore \sum_{i=1}^{20} \left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} \right)^3 = \frac{k}{21}$$

$$\therefore k = 100$$

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Q3

(1) Third term of $(1+x^{\log_2 x})^5 = {}^5C_2 (x^{\log_2 x})^{5-3}$

$$= {}^5C_2 (x^{\log_2 x})^2$$

$$\text{Given, } {}^5C_2 (x^{\log_2 x})^2 = 2560$$

$$\Rightarrow (x^{\log_2 x})^2 = 256 = (\pm 16)^2$$

$$\Rightarrow x^{\log_2 x} = 16 \text{ or } x^{\log_2 x} = -16 \text{ (rejected)}$$

$$\Rightarrow x^{\log_2 x} = 16 \Rightarrow \log_2 x \log_2 x = \log_2 16 = 4$$

$$\Rightarrow \log_2 x = \pm 2 \Rightarrow x = 2^2 \text{ or } 2^{-2}$$

$$\Rightarrow x = 4 \text{ or } \frac{1}{4}$$

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Q4

Binomial Theorem

(1) Since, coefficient of x^2 in the expression $x^2 \left(\sqrt{x} + \frac{\lambda}{x^2} \right)^{10}$ is a constant term, then

$$\text{Coefficient of } x^2 \text{ in } x^2 \left(\sqrt{x} + \frac{\lambda}{x^2} \right)^{10}$$

$$= \text{co-efficient of constant term in } \left(\sqrt{x} + \frac{\lambda}{x^2} \right)^{10}$$

$$\text{General term in } \left(\sqrt{x} + \frac{\lambda}{x^2} \right)^{10} = {}^{10}C_r (\sqrt{x})^{10-r} \left(\frac{\lambda}{x^2} \right)^r$$

$$= {}^{10}C_r (x)^{\frac{10-r}{2}-2r} \cdot \lambda^2$$

Then, for constant term,

$$\frac{10-r}{2} - 2r = 0 \Rightarrow r = 2$$

$$\text{Co-efficient is } x^2 \text{ in expression} = {}^{10}C_2 \lambda^2 = 720$$

$$\Rightarrow \lambda^2 = \frac{720}{5 \times 9} = 16$$

$$\lambda = 4$$

Hence, required value of λ is 4.

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Q5

$$\begin{aligned} (4) \quad \sum_{r=0}^{25} \left({}^{50}C_r \cdot {}^{50-r}C_{25-r} \right) &= \sum_{r=0}^{25} \left(\frac{|50|}{|50-r| \cdot |r|} \cdot \frac{|50-r|}{|25| \cdot |25-r|} \right) \\ &= \sum_{r=0}^{25} \left(\frac{|50|}{|25|} \times \frac{1}{|25|} \times \left(\frac{|25|}{|25-r| \cdot |r|} \right) \right) \\ &= {}^{50}C_{25} \sum_{r=0}^{25} {}^{25}C_r = {}^{50}C_{25} (2^{25}) \end{aligned}$$

Then, by comparison, $K = 2^{25}$

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Q6

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(2) Consider the expression ${}^{20}C_r \cdot {}^{20}C_0 + {}^{20}C_{r-1} \cdot {}^{20}C_1 + {}^{20}C_{r-2} \cdot {}^{20}C_2 + \dots + {}^{20}C_0 \cdot {}^{20}C_r$

For maximum value of above expression r should be equal to 20.

$$\begin{aligned} \text{as } {}^{20}C_{20} \cdot {}^{20}C_0 + {}^{20}C_{19} \cdot {}^{20}C_1 + \dots + {}^{20}C_{20} \cdot {}^{20}C_0 \\ = ({}^{20}C_0)^2 + ({}^{20}C_1)^2 + \dots + ({}^{20}C_{20})^2 = {}^{40}C_{20} \end{aligned}$$

Which is the maximum value of the expression,

So, $r = 20$.

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Q7

(1) Middle Term, $\left(\frac{n}{2} + 1 \right)^{\text{th}}$ term in the binomial

expansion of $\left(\frac{x^3}{3} + \frac{3}{x} \right)^8$ is,

$$T_{4+1} = {}^8C_4 \left(\frac{x^3}{3} \right)^4 \left(\frac{3}{x} \right)^4 = 5670$$

$$\Rightarrow \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2} \times x^{12-4} = 5670$$

$$\Rightarrow x^8 = 81$$

$$\Rightarrow x^8 - 81 = 0$$

$\therefore$ sum of all values of x = sum of roots of equation $(x^8 - 81 = 0)$.

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Q8

Binomial Theorem

$$(4) S_n = \left(\frac{1-q^{n+1}}{1-q} \right), T_n = \frac{1 - \left(\frac{q+1}{2} \right)^{n+1}}{1 - \left(\frac{q+1}{2} \right)}$$

$$\Rightarrow T_{100} = \frac{1 - \left(\frac{q+1}{2} \right)^{101}}{1 - \left(\frac{q+1}{2} \right)}$$

$$S_n = \frac{1}{1-q} - \frac{q^{n+1}}{1-q}, T_{100} = \frac{2^{101} - (q+1)^{101}}{2^{100}(1-q)}$$

$$\text{Now, } {}^{101}C_1 + {}^{101}C_2 S_1 + {}^{101}C_3 S_2 + \dots + {}^{101}C_{101} S_{100}$$

$$= \left(\frac{1}{1-q} \right) ({}^{101}C_2 + \dots + {}^{101}C_{101})$$

$$- \frac{1}{1-q} ({}^{101}C_2 q^2 + {}^{101}C_3 q^3 + \dots + {}^{101}C_{101} q^{101}) + 101$$

$$= \frac{1}{1-q} (2^{101} - 1 - 101) - \left(\frac{1}{1-q} \right) ((1+q)^{101} - 1 - {}^{101}C_1 q) + 101$$

$$= \frac{1}{1-q} [2^{101} - 102 - (1+q)^{101} + 1 + 101q] + 101$$

$$= \frac{1}{1-q} [2^{101} - 101 + 101q - (1+q)^{101}] + 101$$

$$= \left(\frac{1}{1-q} \right) [2^{101} - (1+q)^{101}] = 2^{100} T_{100}$$

Hence, by comparison $\alpha = 2^{100}$

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Q9

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$$(3) (x+10)^{50} + (x-10)^{50}$$

$$= a_0 + a_1 x + a_2 x^2 + \dots + a_{50} x^{50}$$

$$\therefore a_0 + a_1 x + a_2 x^2 + \dots + a_{50} x^{50}$$

$$= 2({}^{50}C_0 x^{50} + {}^{50}C_2 x^{48} \cdot 10^2 + {}^{50}C_4 x^{46} \cdot 10^4 + \dots)$$

$$\therefore a_0 = 2 \cdot {}^{50}C_{50} 10^{50}$$

$$a_2 = 2 \cdot {}^{50}C_2 \cdot 10^{48}$$

$$\therefore \frac{a_2}{a_0} = \frac{{}^{50}C_2 \times 10^{48}}{{}^{50}C_{50} 10^{50}}$$

$$= \frac{50 \times 49}{2 \times 100} = \frac{49}{4}$$

$$= 12.25$$

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Q10

Binomial Theorem

$$(3) \left(2^{\frac{1}{3}} + \frac{1}{2(3)^{\frac{1}{3}}} \right)^{10} = {}^{10}C_0 \left(2^{\frac{1}{3}} \right)^0 \left(\frac{1}{2(3)^{\frac{1}{3}}} \right)^{10} +$$

$$\dots + {}^{10}C_{10} \left(2^{\frac{1}{3}} \right)^{10} \left(\frac{1}{2(3)^{\frac{1}{3}}} \right)^0$$

$$5^{\text{th}} \text{ term from beginning } T_5 = {}^{10}C_4 \left(2^{\frac{1}{3}} \right)^6 \frac{1}{\left(2.3^{\frac{1}{3}} \right)^4}$$

$$\text{and } 5^{\text{th}} \text{ term from end } T_{11-5+1} = {}^{10}C_6 \left(2^{\frac{1}{3}} \right)^4 \left(\frac{1}{2.3^{\frac{1}{3}}} \right)^6$$

$$\therefore T_5 : T_7 =$$

$${}^{10}C_4 \left(2^{\frac{1}{3}} \right)^6 \left(\frac{1}{2.3^{\frac{1}{3}}} \right)^4 : {}^{10}C_6 \left(2^{\frac{1}{3}} \right)^4 \left(\frac{1}{2.3^{\frac{1}{3}}} \right)^6$$

$$= \left(2^{\frac{1}{3}} \right)^2 : \left(\frac{1}{2.3^{\frac{1}{3}}} \right)^2$$

$$= \frac{2^{\frac{2}{3}} \cdot 2^2 \cdot 3^{\frac{2}{3}}}{1} = 4(6)^{\frac{2}{3}} : 1 = 4.(36)^{\frac{1}{3}} : 1$$

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Q11

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(4) Let the general term of the expansion

$$T_{r+1} = {}^{60}C_r \left(7^{\frac{1}{5}} \right)^{60-r} \left(-3^{\frac{1}{10}} \right)^r$$

$$= {}^{60}C_r \cdot (7)^{12-\frac{r}{5}} (-1)^r \cdot (3)^{\frac{r}{10}}$$

Then, for getting rational terms, r should be multiple of L.C.M. of (5, 10)

Then, r can be 0, 10, 20, 30, 40, 50, 60.

Since, total number of terms = 61

Hence, total irrational terms = $61 - 7 = 54$

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Q12

$$(4) (x + \sqrt{x^3 - 1})^6 + (x - \sqrt{x^3 - 1})^6$$

$$= 2[{}^6C_0 x^6 + {}^6C_2 x^4 (x^3 - 1) + {}^6C_4 x^2 (x^3 - 1)^2 + {}^6C_6 (x^3 - 1)^3]$$

$$= 2[x^6 + 15x^7 - 15x^4 + 15x^8 - 30x^5 + 15x^2 + x^9 - 3x^6 + 3x^3 - 1]$$

Hence, the sum of coefficients of even powers of

$$x = 2[1 - 15 + 15 + 15 - 3 - 1] = 24$$

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Q13

$$(2) 2.{}^{20}C_0 + 5.{}^{20}C_1 + 8.{}^{20}C_2 + \dots + 62.{}^{20}C_{20}$$

$$= \sum_{r=0}^{20} (3r+2) {}^{20}C_r = 3 \sum_{r=0}^{20} r. {}^{20}C_r + 2 \sum_{r=0}^{20} {}^{20}C_r$$

$$= 60 \sum_{r=1}^{20} {}^{19}C_{r-1} + 2 \sum_{r=0}^{20} {}^{20}C_r$$

$$= 60 \times 2^{19} + 2 \times 2^{20} = 2^{21} [15 + 1] = 2^{25}$$

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Q14

Binomial Theorem

(2) $\therefore$ fourth term is equal to 200.

$$T_4 = {}^6C_3 \left(\sqrt{x \left(\frac{1}{1+\log_{10} x} \right)} \right)^3 \left(\frac{1}{x^{12}} \right)^3 = 200$$

$$\Rightarrow 20x^{\frac{3}{2(1+\log_{10} x)}} \cdot \frac{1}{x^4} = 200$$

$$\frac{1}{x^4} + \frac{3}{2(1+\log_{10} x)} = 10$$

Taking $\log_{10}$ on both sides and putting $\log_{10} x = t$

$$\left(\frac{1}{4} + \frac{3}{2(1+t)} \right) t = 1 \Rightarrow t^2 + 3t - 4 = 0$$

$$\Rightarrow t^2 + 4t - t - 4 = 0 \Rightarrow t(t+4) - 1(t+4) = 0$$

$$\Rightarrow t = 1 \text{ or } t = -4$$

$$\log_{10} x = 1 \Rightarrow x = 10$$

$$\text{or } \log_{10} x = -4 \Rightarrow x = 10^{-4}$$

According to the question $x > 1$, $\therefore x = 10$.

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Q15

(2) $\because T_4 = 20 \times 8^7$

$$\Rightarrow {}^6C_3 \left(\frac{2}{x} \right)^3 \times (x^{\log_8 x})^3 = 20 \times 8^7$$

$$\Rightarrow 8 \times 20 \times \left(\frac{x^{\log_8 x}}{x} \right)^3 = 20 \times 8^7$$

$$\Rightarrow \frac{x^{\log_8 x}}{x} = 64$$

Now, take $\log_8$ on both sides, we get

$$(\log_8 x)^2 - (\log_8 x) = 2$$

$$\Rightarrow \log_8 x = -1 \quad \text{or } \log_8 x = 2$$

$$\Rightarrow x = \frac{1}{8} \quad \text{or } x = 8^2$$

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Q16

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(2) Given ${}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 2 : 15 : 70$

$$\Rightarrow \frac{{}^nC_{r-1}}{{}^nC_r} = \frac{2}{15} \text{ and } \frac{{}^nC_r}{{}^nC_{r+1}} = \frac{15}{70}$$

$$\Rightarrow \frac{r}{n-r+1} = \frac{2}{15} \text{ and } \frac{r+1}{n-r} = \frac{3}{14}$$

$$\Rightarrow 17r = 2n + 2 \text{ and } 17r = 3n - 14$$

$$\text{i.e., } 2n + 2 = 3n - 14 \Rightarrow n = 16 \text{ \& } r = 2$$

$$\therefore \text{Average} = \frac{{}^{16}C_1 + {}^{16}C_2 + {}^{16}C_3}{3} = \frac{16 + 120 + 560}{3}$$

$$= \frac{696}{3} = 232$$

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Q17

(3) Given expression is $(1 + ax + bx^2)(1 - 3x)^{15}$

Co-efficient of $x^2 = 0$

$$\Rightarrow {}^{15}C_2(-3)^2 + a \cdot {}^{15}C_1(-3) + b \cdot {}^{15}C_0 = 0$$

$$\Rightarrow \frac{15 \times 14}{2} \times 9 - 15 \times 3a + b = 0$$

$$\Rightarrow 945 - 45a + b = 0 \quad \dots(i)$$

Now, co-efficient of $x^3 = 0$

$$\Rightarrow {}^{15}C_3(-3)^3 + a \cdot {}^{15}C_2(-3)^2 + b \cdot {}^{15}C_1(-3) = 0$$

$$\Rightarrow \frac{15 \times 14 \times 13}{3 \times 2} \times (-3 \times 3 \times 3) + a \times \frac{15 \times 14 \times 9}{2}$$

$$- b \times 3 \times 15 = 0$$

$$\Rightarrow 15 \times 3 [-3 \times 7 \times 13 + a \times 7 \times 3 - b] = 0$$

$$\Rightarrow 21a - b = 273 \quad \dots(ii)$$

From (i) and (ii), we get,

$$a = 28, b = 315 \Rightarrow (a, b) \equiv (28, 315)$$

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Q18

(1) $\left(x^2 + \frac{1}{x^2} \right)^n$

$$\text{General term } T_{r+1} = {}^nC_r (x^2)^{n-r} \left(\frac{1}{x^2} \right)^r = {}^nC_r \cdot x^{2n-5r}$$

To find coefficient of x , $2n - 5r = 1$

$$\text{Given } {}^nC_r = {}^nC_{23} \Rightarrow r = 23 \text{ or } n - r = 23$$

$$\therefore n = 58 \text{ or } n = 38$$

Minimum value is $n = 38$

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Binomial Theorem

Q19

(1) Given expression,

$$\begin{aligned}(1-x)^{10} (1+x+x^2)^9 (1+x) &= (1-x^3)^9 (1-x^2) \\ &= (1-x^3)^9 - x^2(1-x^3)^9 \\ \Rightarrow \text{Coefficient of } x^{18} \text{ in } (1-x^3)^9 &= \text{coeff. of } x^{16} \text{ in } (1-x^3)^9 \\ &= {}^9C_6 - 0 = \frac{9!}{6!3!} = \frac{7 \times 8 \times 9}{6} = 84\end{aligned}$$

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Q20

(2) Given, ${}^{20}C_1 + 2^2 \cdot {}^{20}C_2 + 3^2 \cdot {}^{20}C_3 + \dots + 20^2 \cdot {}^{20}C_{20} = A(2^\beta)$

Taking L.H.S.,

$$\begin{aligned}&= \sum_{r=1}^{20} r^2 \cdot {}^{20}C_r = 20 \sum_{r=1}^{20} r \cdot {}^{19}C_{r-1} \\ &= 20 \left[\sum_{r=1}^{20} (r-1) {}^{19}C_{r-1} + \sum_{r=1}^{20} {}^{19}C_{r-1} \right] \\ &= 20 \left[19 \sum_{r=2}^{20} {}^{18}C_{r-2} + 2^{19} \right] = 20[19 \cdot 2^{18} + 2^{19}] \\ &= 420 \times 2^{18}\end{aligned}$$

Now, compare it with R.H.S., $A = 420$ and $\beta = 18$

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Q21

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(4) Given expression is,

$$\begin{aligned}&\left(\frac{1}{60} - \frac{x^8}{81} \right) \left(2x^2 - \frac{3}{x^2} \right)^6 \\ &= \frac{1}{60} \left(2x^2 - \frac{3}{x^2} \right)^6 - \frac{x^8}{81} \left(2x^2 - \frac{3}{x^2} \right)^6\end{aligned}$$

Term independent of x ,

$$\begin{aligned}&= \text{Coefficient of } x^0 \text{ in } \frac{1}{60} \left(2x^2 - \frac{3}{x^2} \right)^6 - \frac{1}{81} \cdot \\ &\text{coefficient of } x^{-8} \text{ in } \left(2x^2 - \frac{3}{x^2} \right)^6 \\ &= \frac{-1}{60} {}^6C_3 (2)^3 (3)^3 + \frac{1}{81} {}^6C_5 (2) (3)^5 \\ &= -72 + 36 = -36\end{aligned}$$

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